Name: Yung Gadi and the Dusty Bunch **Date:** 2/9/24 **Class:** Eng Systems 1

MyDesign Portfolio

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1 (HIGH Priority)

The solution must prevent sawdust from entering the classroom and building up in the computers.

From Mr. L's testimony, a direct issue from the production of sawdust in the workshop is its escape into the classroom, flowing through the air and damaging classroom materials like its computers and terminals. An article by Vanguard Cleaning details the issues of having dust in computers, including the blockage of air vents and fans which prevents heat from dissipating. This causes inefficient cooling, which lowers computer performance, longevity, or even malfunction in the worst cases.

Vanguard Cleaning Systems of the Southern Valley. "What Dust Does to Computer Equipment." *Vanguard Cleaning Systems*, 1 Apr. 2019, www.vanguardsv.com/2019/03/what-dust-does-to-computer-equipment/#:~:text=Computer%20and%20server%20hardware%20is,solder%2C%20preventing%20heat%20from%20dissipating.

Name: Yung Gadi and the Dusty Bunch **Date:** 2/9/24 **Class:** Eng Systems 1

Design Requirement #2 (HIGH Priority)

The solution should be intuitive to learn, and easy to remember.

Mr. L, our primary stakeholder, complains that the current ventilation system takes time, effort, and awareness to use effectively and prevent the spread of dust, especially by students who often forget to pay attention to use it correctly. The current ventilation system requires the user to check 5 separate locations to ensure efficiency, so the solution should be intuitive to use, and either minimal effort to remember, or to incorporate reminders in order to ensure awareness of its setup. Visual reminders are key to awareness; In a study published in the National Library of Medicine, researchers proved that on average, nurses in a hospital were more likely to complete tasks if they were accompanied by a visual reminder or checklist.

> Grundgeiger T, Sanderson PM, Orihuela CB, Thompson A, MacDougall HG, Nunnink L, Venkatesh B. Prospective memory in the ICU: the effect of visual cues on task execution in a representative simulation. Ergonomics. 2013;56(4):579-89. doi: 10.1080/00140139.2013.765604. Epub 2013 Mar 21. PMID: 23514201.

Design Requirement #3 (HIGH Priority)

The solution should be able to prevent the buildup of sawdust on equipment.

One issue of the current system in place to tackle the problem of sawdust in the workshop, being the pipe-based ventilation system, is that dust continues to accumulate on the woodworking equipment, often in places unreachable by the suction of the vents. Mr. L complains that oftentimes, an external “shop-vac” must be used to clear the machinery of dust. A solution must be able to clear the sawdust from the machinery, or localize it where it is produced, so it can be removed efficiently.

Name: Yung Gadi and the Dusty Bunch **Date:** 2/9/24 **Class:** Eng Systems 1

Design Requirement #4 (HIGH Priority)

The solution should be able to continue working with minimal repair and/or upkeep.

A good solution for any problem should be able to continue functioning without constant supervision by its stakeholders or its users, so should be the same case for a solution regarding dust. When a good solution lacks proper longevity, and falls to disrepair quickly after being effective, then it can sometimes be worse than not having it in place at all, as further resources must be sank into its upkeep. Mr L. adds that the currently solutions qualify for this requirement well, and remain robust and functional years after their installation.

Design Requirement #5 (HIGH Priority)

The solution should be effective at removing built up dust on workshop user's clothes and hair.

A mildly contentious requirement with Mr L, the issue of tracking dust into the classroom extends to the clothing and hair of students in the workshop. Often when tools are in use, and dust is flying through the air, then it can attach itself into the fabric of shirts or pants, or getting caught in people's hair. One especially bad case that Mr. L brought up was when a student was using the sander, and after a while when they came out, their face was colored pale, covered in sawdust. Now, a solution that solved this problem might be a but further from the initial boundaries, as it steps into the territories of yet another issue, but ideas and concepts should still be considered.

Design Requirement #6 (LOW Priority)

The solution should prevent sawdust from entering the lungs of students and faculty.

The issue of sawdust entering the lungs of people in the workshop, both student and faculty alike, is indeed a problem worth considering. However, since humans are resilient creatures, and do not get damaged by smaller amounts of dust easily, then this issue can remain a low priority. Mr. L confirms as such, as no prior student has faced serious harm or ailment from inhaling the sawdust produced in the workshop.

Name: Yung Gadi and the Dusty Bunch **Date:** 2/9/24 **Class:** Eng Systems 1

Design Requirement #7 (LOW Priority)

The solution should be visually appealing and pleasing to the eyes.

A very low priority. As the solution will be implemented in a workshop, where the areas will often be messy with tools, works in progress, equipment and yes, sawdust, then visual aesthetics will not need to be prioritized, especially not over functionality or practical purpose. Mr. L agrees, and doesn't care if its pretty, only if it works.